

Equipment Inspection & Maintenance Report

Report No: MNT-2026-0142
Plant: North Sector Process Unit 3
Inspection Date: 14 March 2026
Inspector: R. Menon, Senior Process Safety Engineer

1. Summary

Routine inspection of pump P-101A and associated pressure relief valve V-204 was carried out on 14 March 2026 following a reported pressure fluctuation logged on 09 March 2026. Vibration readings on P-101A exceeded the baseline threshold by 18%%, consistent with early-stage bearing wear. No gas leakage was detected during the inspection window.

2. Findings

Operating pressure on line PL-22 was recorded at 8.7 bar against a rated maximum of 10.2 bar. Bearing housing temperature on P-101A reached 71 degrees Celsius, above the 65 degree Celsius normal operating range specified in the OEM manual for this pump model. Relief valve V-204 set pressure was verified at 9.8 bar, within tolerance.

3. Regulatory Cross-Reference

This inspection was conducted in accordance with OISD-STD-118 (Layouts for Oil and Gas Installations) safe-distance provisions and Factory Act Sec. 21 (Safety of Building and Machinery) periodic inspection requirements. No deviations from mandated inspection intervals were identified.

4. Prior Related Incidents

Maintenance log MNT-2025-0871 (dated 22 August 2025) recorded a similar vibration anomaly on pump P-101A, attributed at the time to a loose foundation bolt and resolved by re-torquing to OEM specification. Given the recurrence pattern, bearing replacement is recommended rather than a further torque adjustment.

5. Recommended Actions

Action	Equipment	Priority	Due Date
Replace bearing assembly	P-101A	High	28 March 2026
Re-verify relief set pressure post-repair	V-204	Medium	30 March 2026
Update OEM service interval log	P-101A	Low	31 March 2026